

METIMMOX-1 Value of Information Analysis

Biomarker-Guided Strategies for Selecting Immunotherapy Candidates in Metastatic MSS/pMMR Colorectal Cancer

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1 Overview

This report presents the results of an Expected Value of Partially Perfect Information (EVPPI) analysis for the METIMMOX-1 decision-analytic model. The analysis evaluates the value of reducing uncertainty in individual model parameters across multiple scenarios with varying willingness-to-pay (WTP) thresholds and nivolumab drug costs.

The Expected Value of Partially Perfect Information (EVPPI) quantifies the value of eliminating uncertainty in specific model parameters. It represents the maximum amount that should be paid for additional research to perfectly inform a particular parameter. EVPI (Expected Value of Perfect Information) represents the value of eliminating all uncertainty, and EVPPI for any parameter cannot exceed total EVPI.

Table 1: Analysis scenario definitions. Higher WTP thresholds (€100,000 and €150,000) are arbitrary exploratory thresholds, though the highest severity adjustment in Norwegian guidelines may fall between these values. The decreased nivolumab cost (€4,641) represents an 80% discount compared to the 40% discount in the base case scenario.

Scenario Name	WTP Threshold	Nivolumab Cost
Base Case	€ 51,000	€13,923
Decreased Nivolumab Cost	€ 51,000	€ 4,641
WTP €100,000	€100,000	€13,923
WTP €100,000 + Decreased Nivolumab	€100,000	€ 4,641
WTP €150,000	€150,000	€13,923
WTP €150,000 + Decreased Nivolumab	€150,000	€ 4,641

2 Base Case Scenario

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €13,923

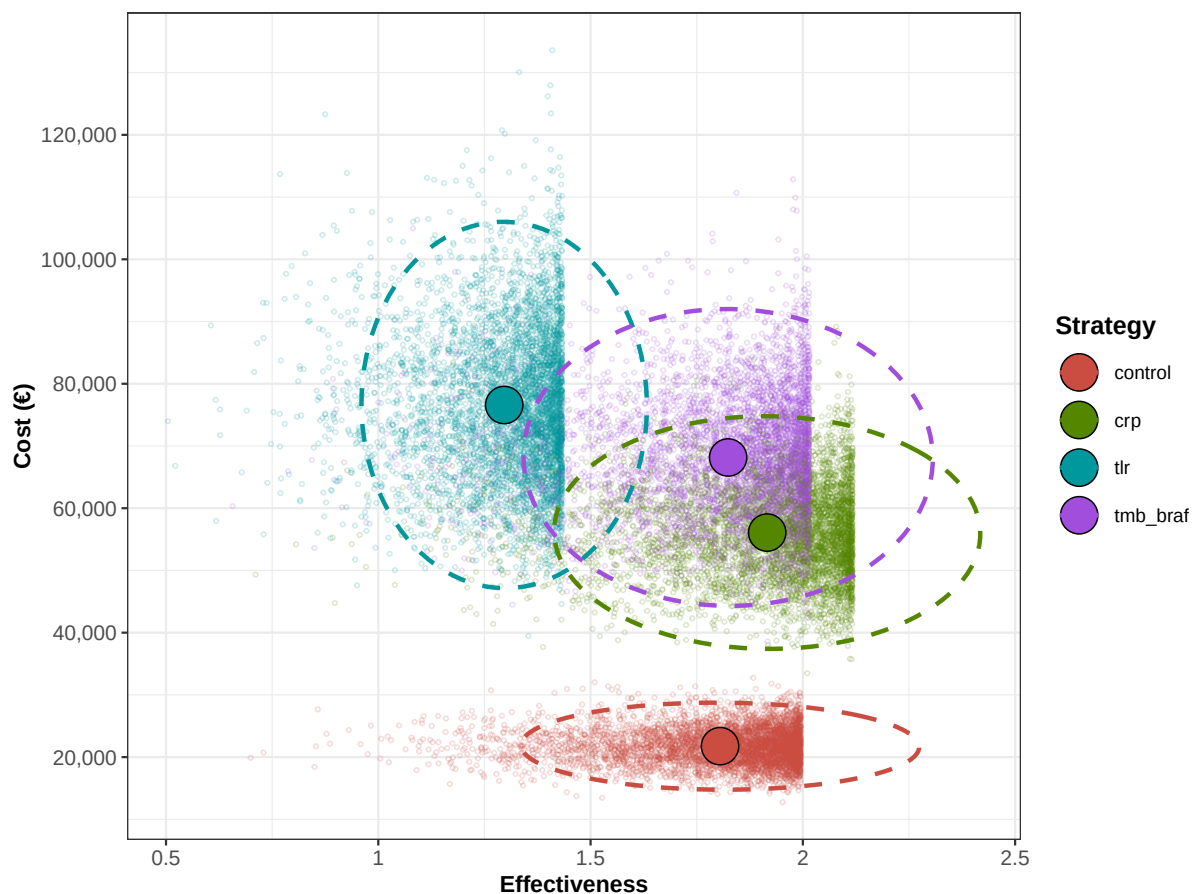


Figure 1: Cost-effectiveness scatter plot (base case)

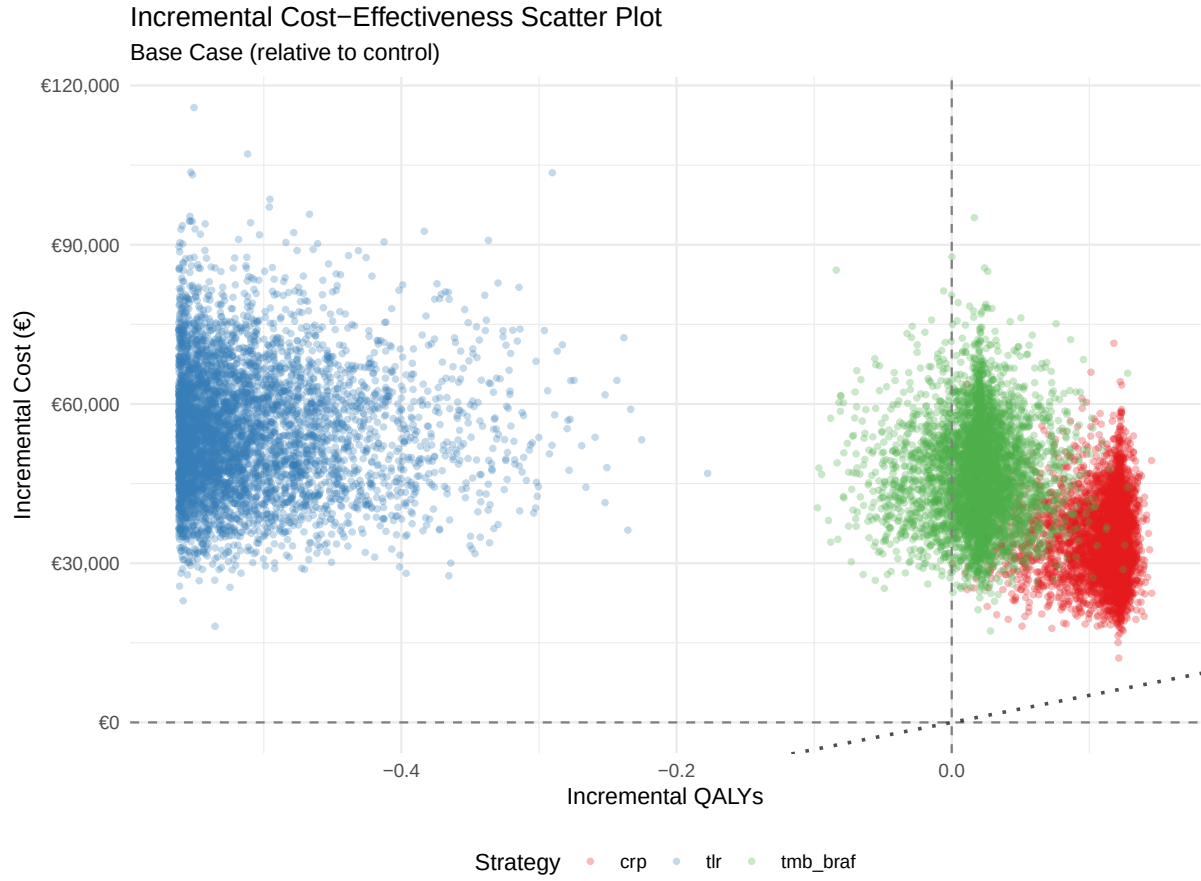


Figure 2: Incremental cost-effectiveness scatter plot relative to control (base case)

Table 2: Cost-effectiveness analysis summary from probabilistic sensitivity analysis (base case)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€21,758.31	€ 2,856.78	1.805	0.192	€ 70,298.68	€10,236.61
crp	€56,086.29	€ 7,638.79	1.916	0.205	€ 41,632.78	€13,019.74
tlr	€76,586.70	€12,026.29	1.297	0.137	€-10,462.25	€13,963.40
tmb_braf	€68,154.60	€ 9,745.35	1.824	0.197	€ 24,872.44	€14,058.86

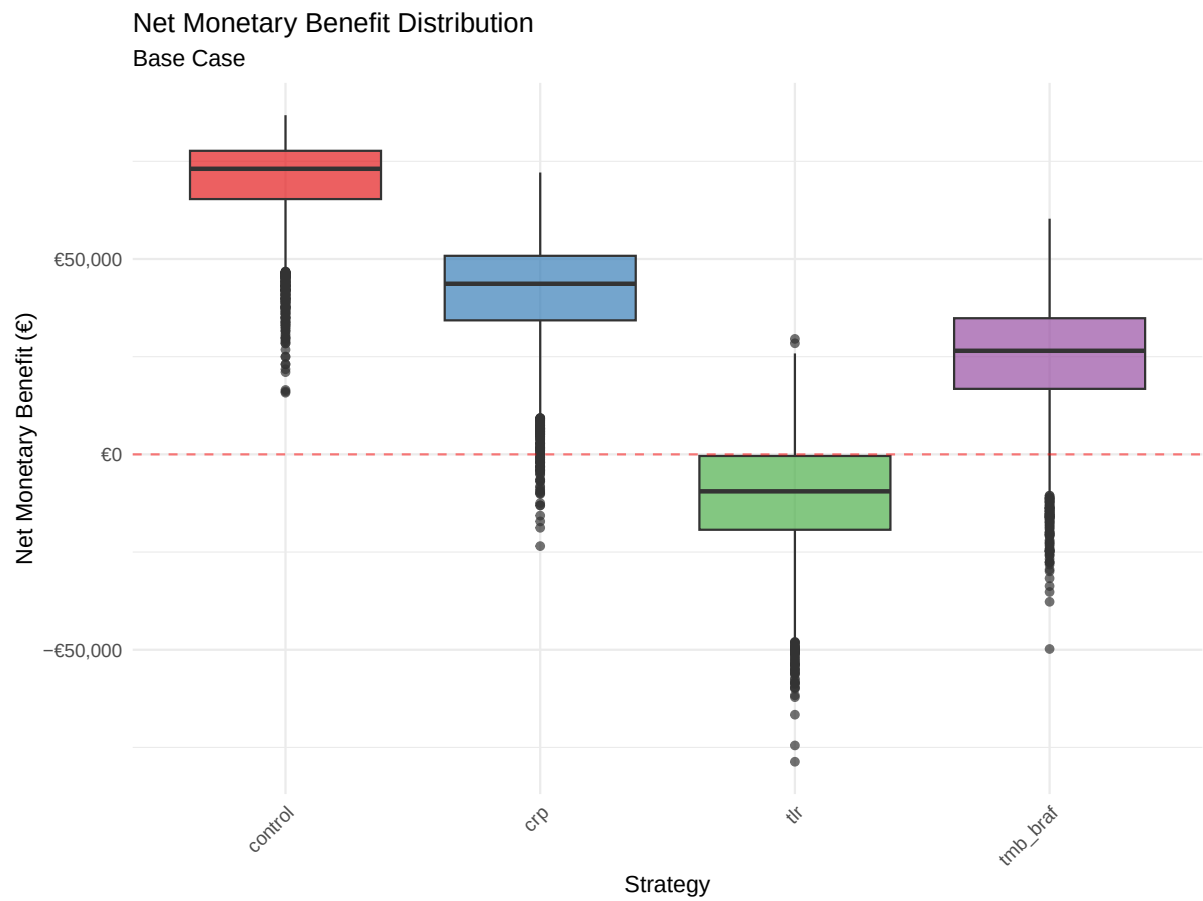


Figure 3: Net Monetary Benefit distribution (base case)

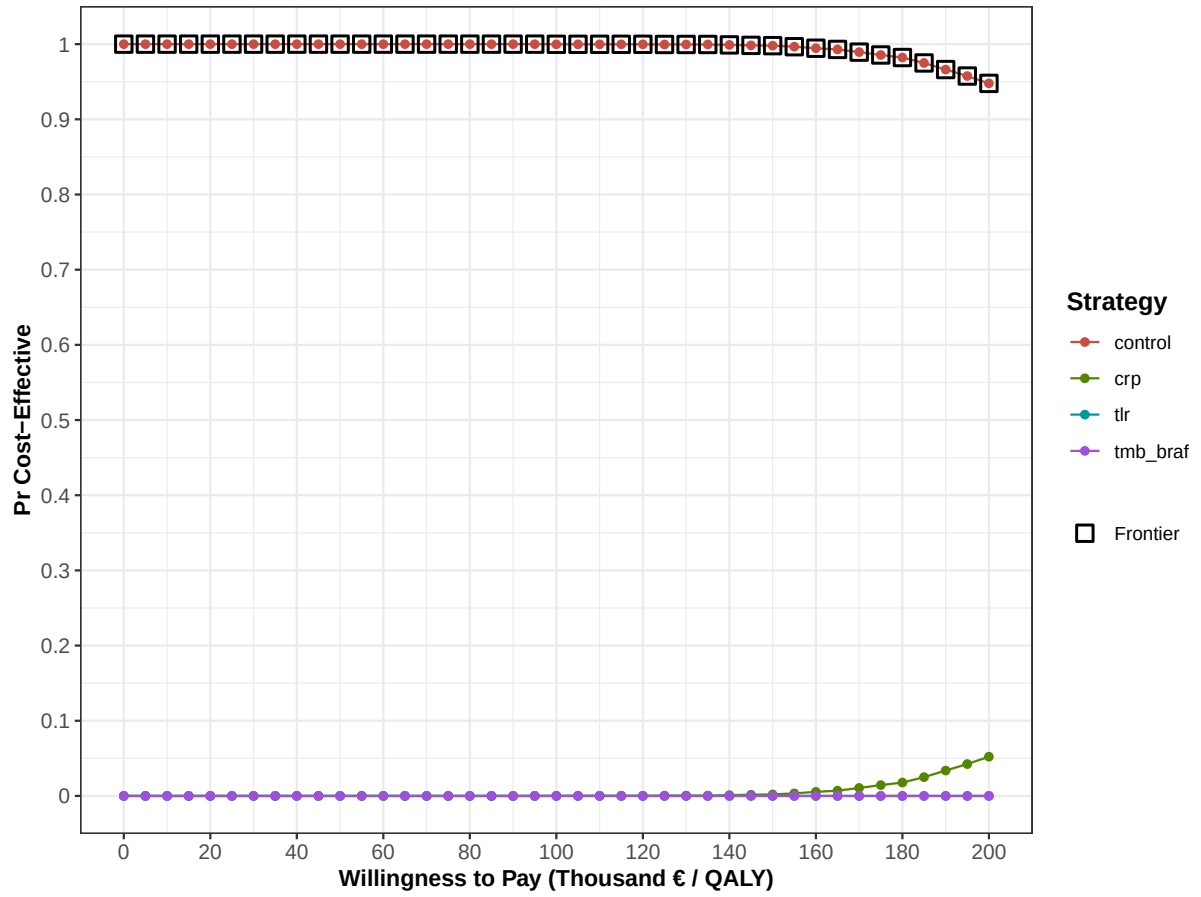


Figure 4: Cost-effectiveness acceptability curves (base case)

Total EVPI: €0

The EVPI is effectively zero, indicating no decision uncertainty at the base case WTP threshold. In this case, one strategy is clearly dominant regardless of parameter uncertainty, making EVPPI calculations less meaningful for the base case scenario.

3 Scenario 1: Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €51,000

Nivolumab Cost: €4,641 (80% discount)

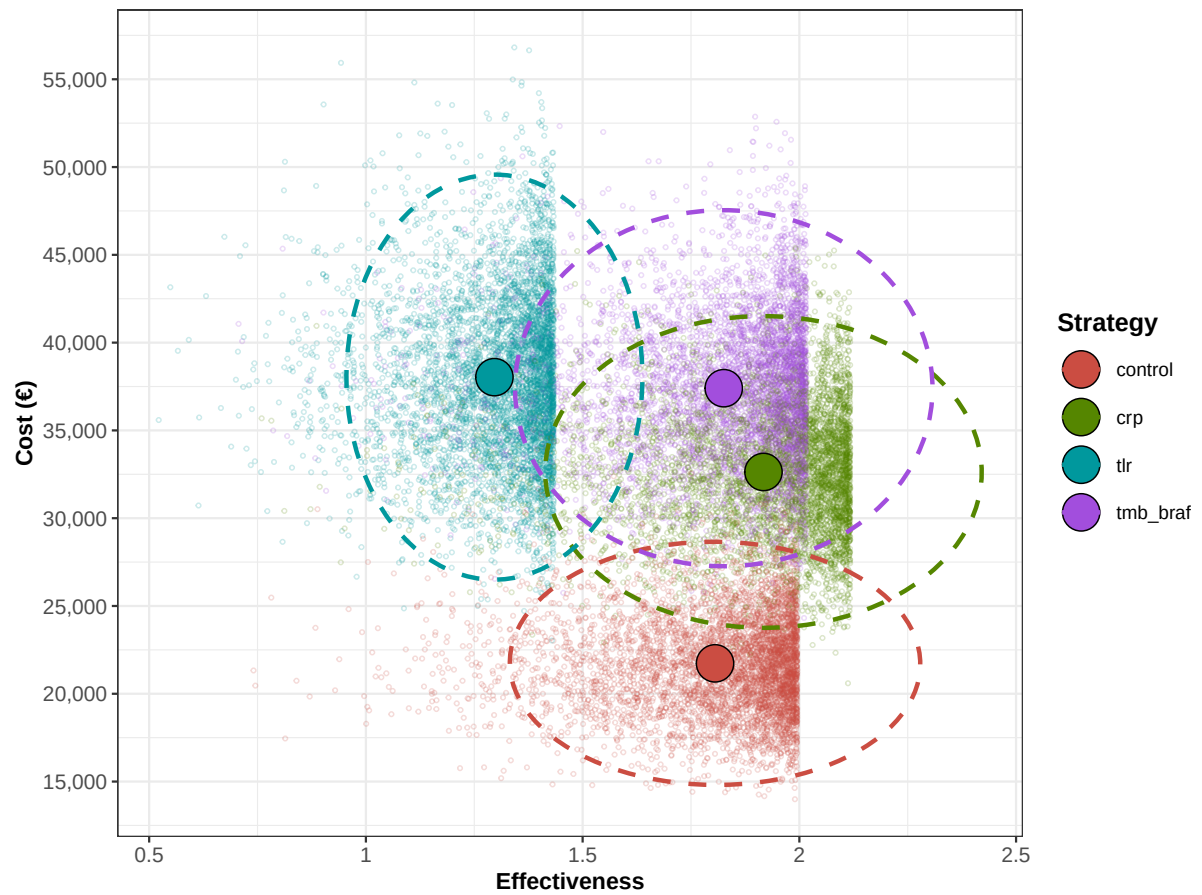


Figure 5: Cost-effectiveness scatter plot (Scenario 1)

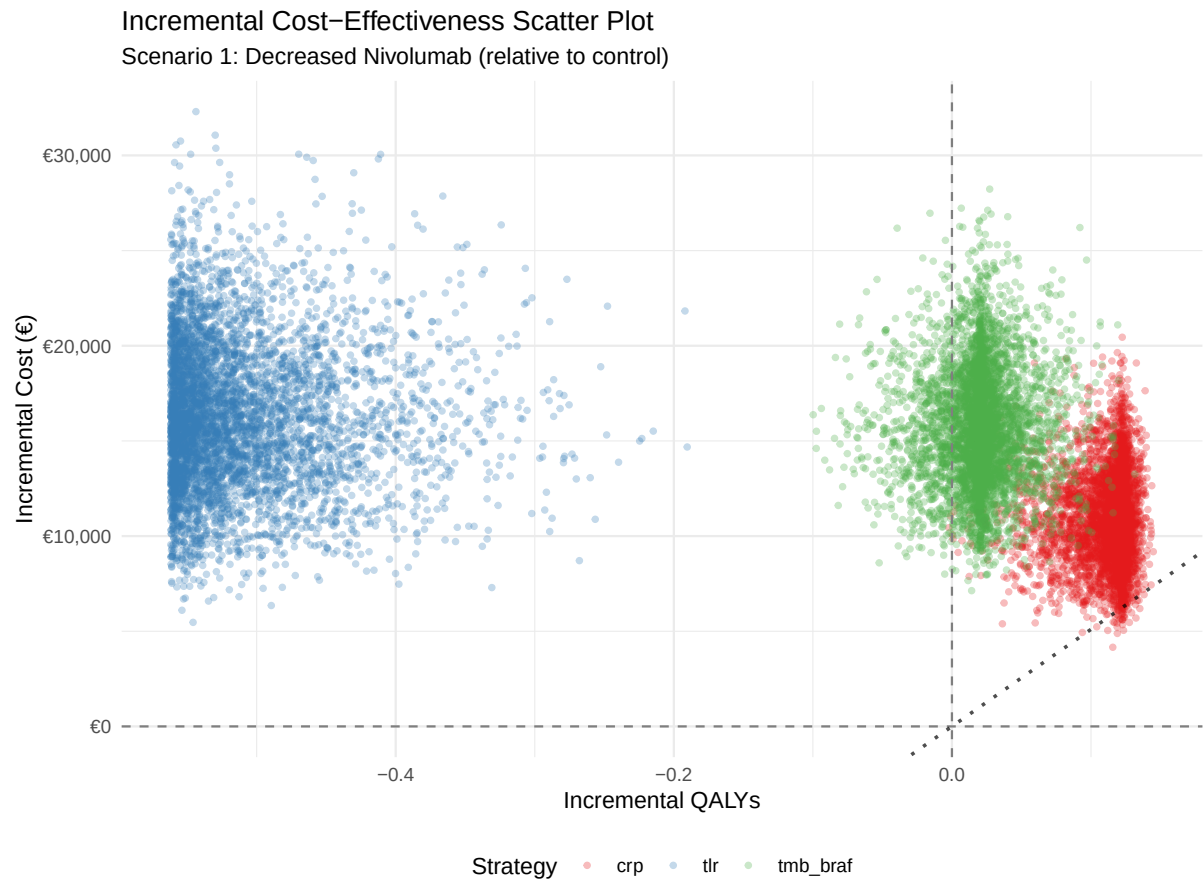


Figure 6: Incremental cost-effectiveness scatter plot relative to control (Scenario 1)

Table 3: Cost-effectiveness analysis summary (Scenario 1)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€21,731.75	€2,829.06	1.806	0.193	€70,349.14	€10,281.86
crp	€32,633.36	€3,627.37	1.917	0.206	€65,145.46	€11,108.27
tlr	€38,034.36	€4,713.70	1.296	0.140	€28,085.79	€ 8,528.75
tmb_braf	€37,404.23	€4,142.88	1.826	0.197	€55,701.23	€10,861.31

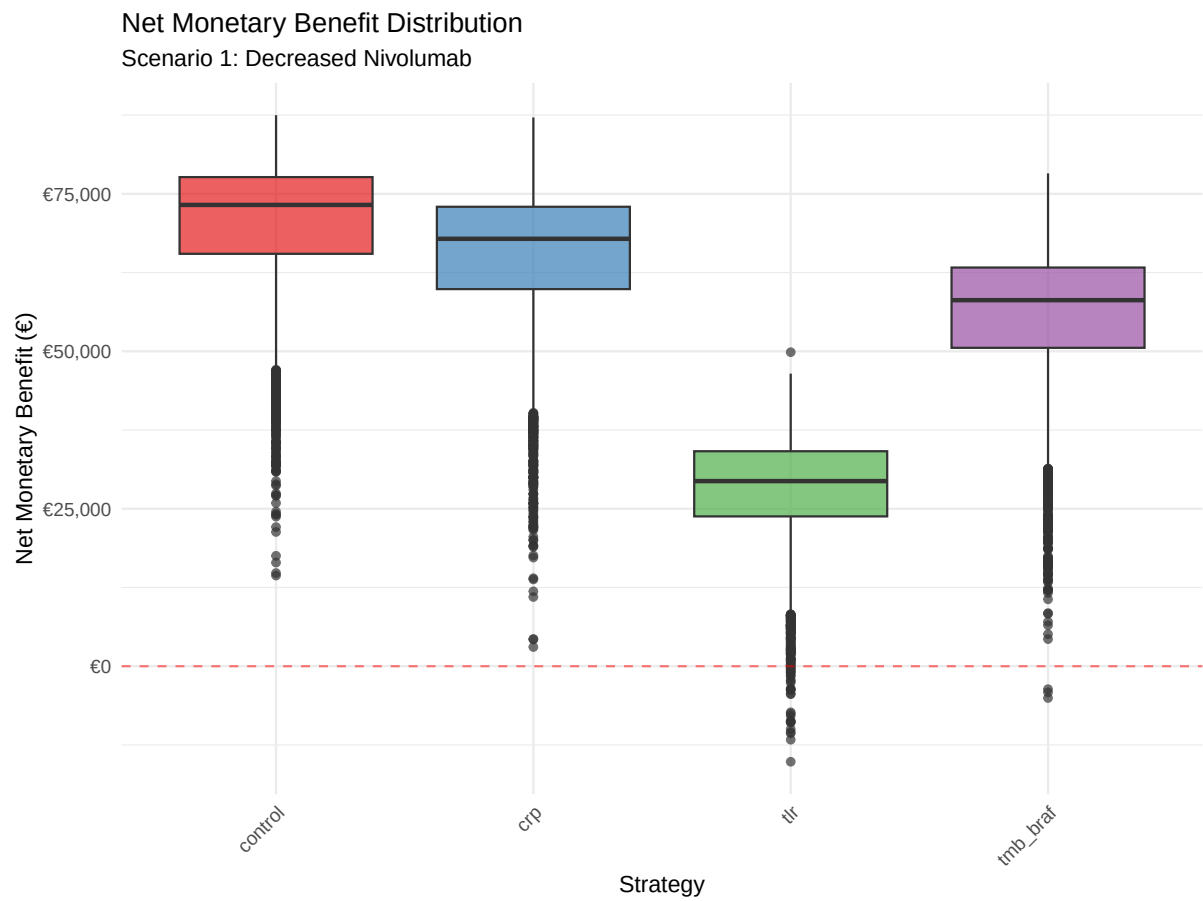


Figure 7: Net Monetary Benefit distribution (Scenario 1)

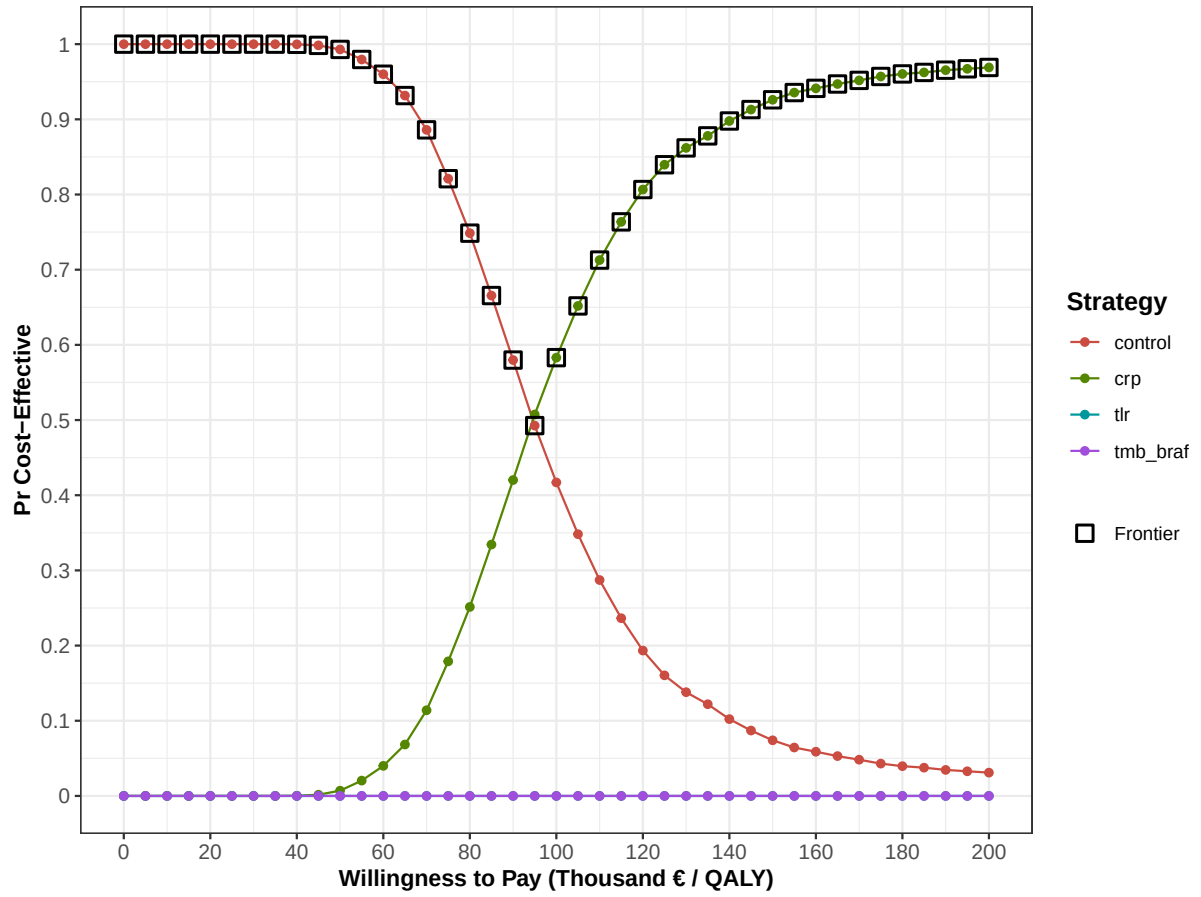


Figure 8: Cost-effectiveness acceptability curves (Scenario 1)

Total EVPI: €4.01

Reducing nivolumab costs creates decision uncertainty, making EVPPI analysis meaningful.

Table 4: EVPPI results: Decreased nivolumab cost

Parameter	EVPI	% of EVPI
c_drug_nivo	€4.01	100%
c_test_CT	€4.01	100%
u_np	€4.01	100%
u_p	€4.01	100%
c_other_last	€4.01	100%
c_other_baseline	€4.01	100%
c_drug_FLOX	€0	0%
c_test_NGS	€0	0%
c_other_visit	€0	0%
c_other_follow	€0	0%

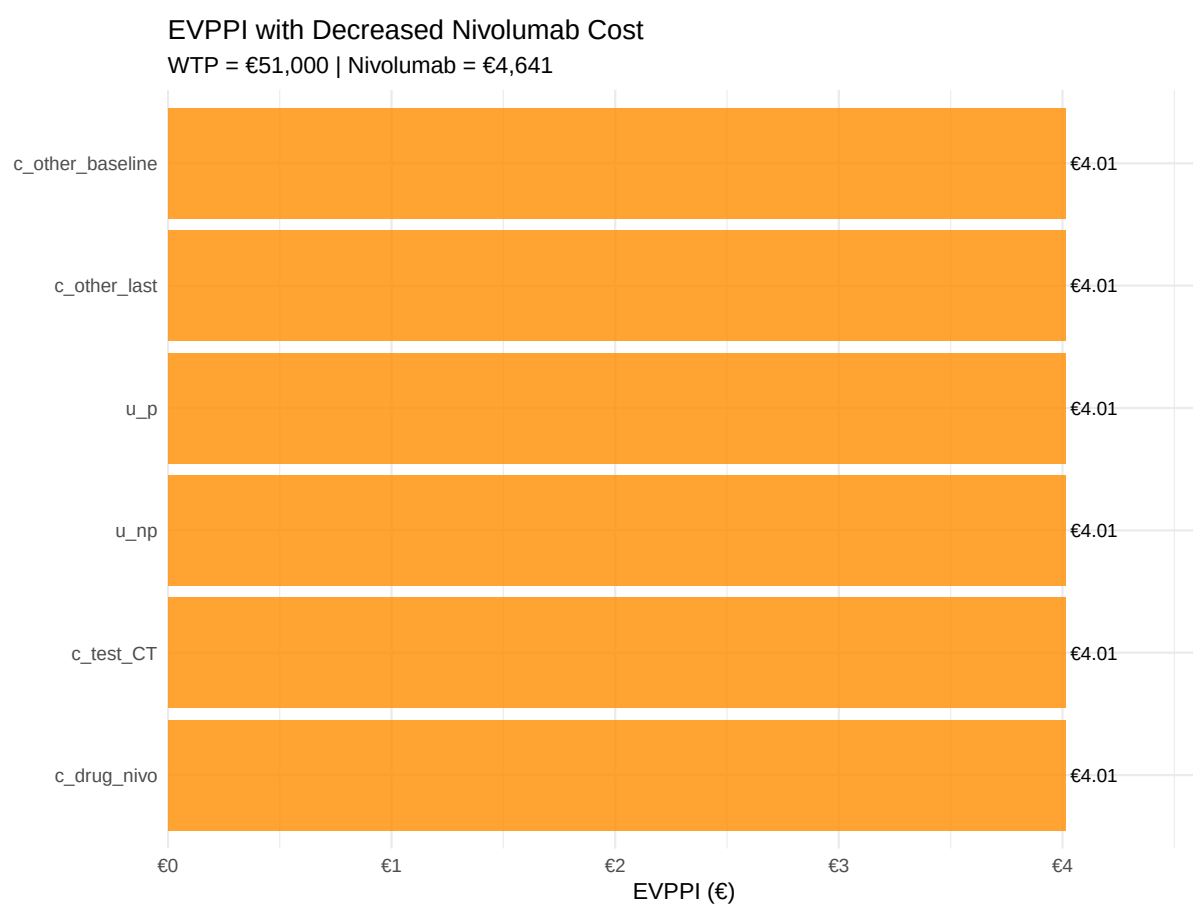


Figure 9: EVPPPI with decreased nivolumab cost

4 Scenario 2: Higher WTP Threshold (€100,000)

Willingness-to-Pay Threshold: €1e+05

Nivolumab Cost: €13,923 (standard)

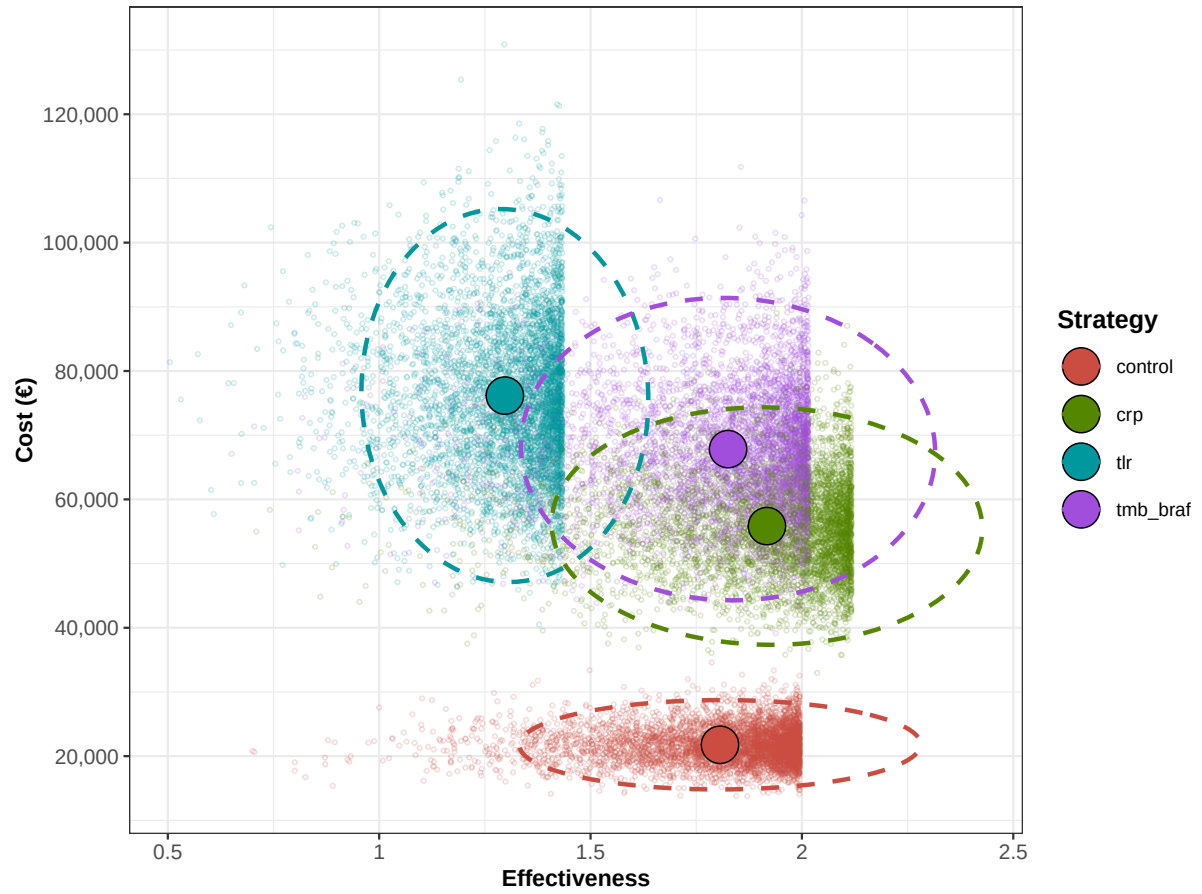


Figure 10: Cost-effectiveness scatter plot (Scenario 2)

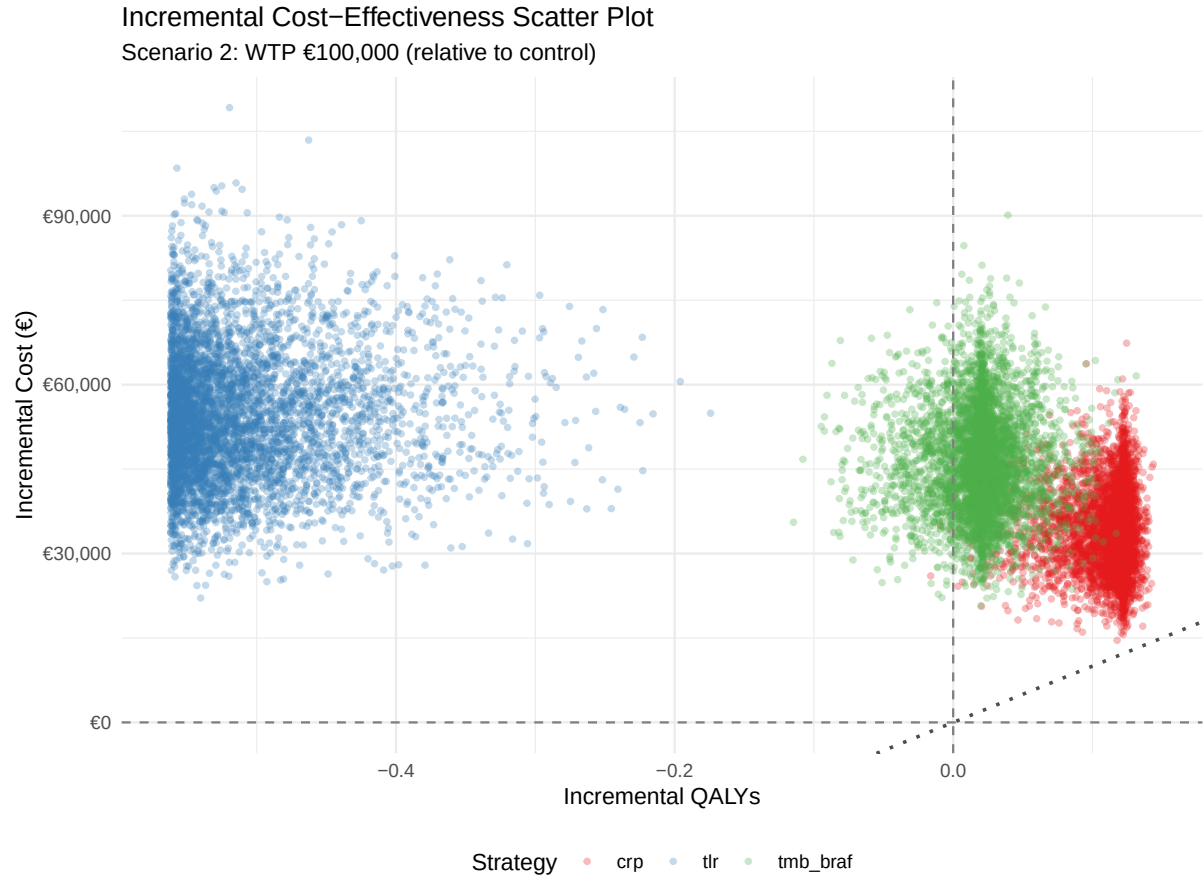


Figure 11: Incremental cost-effectiveness scatter plot relative to control (Scenario 2)

Table 5: Cost-effectiveness analysis summary (Scenario 2)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€21,772.44	€ 2,843.93	1.806	0.194	€158,844.71	€19,575.81
crp	€55,841.03	€ 7,557.27	1.917	0.208	€135,888.56	€22,297.81
tlr	€76,171.90	€11,872.79	1.297	0.139	€ 53,563.13	€18,482.56
tmb_braf	€67,830.11	€ 9,626.76	1.825	0.200	€114,695.59	€22,416.81

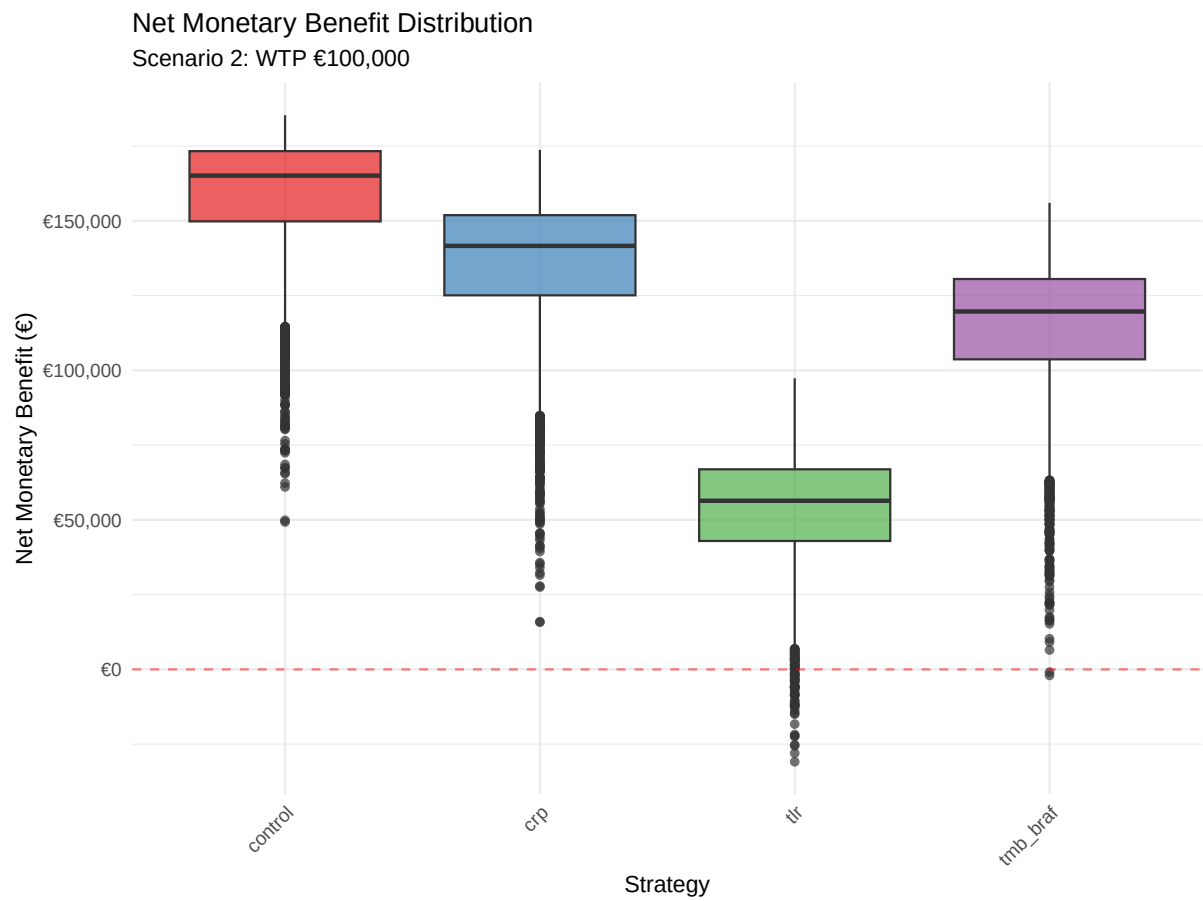


Figure 12: Net Monetary Benefit distribution (Scenario 2)

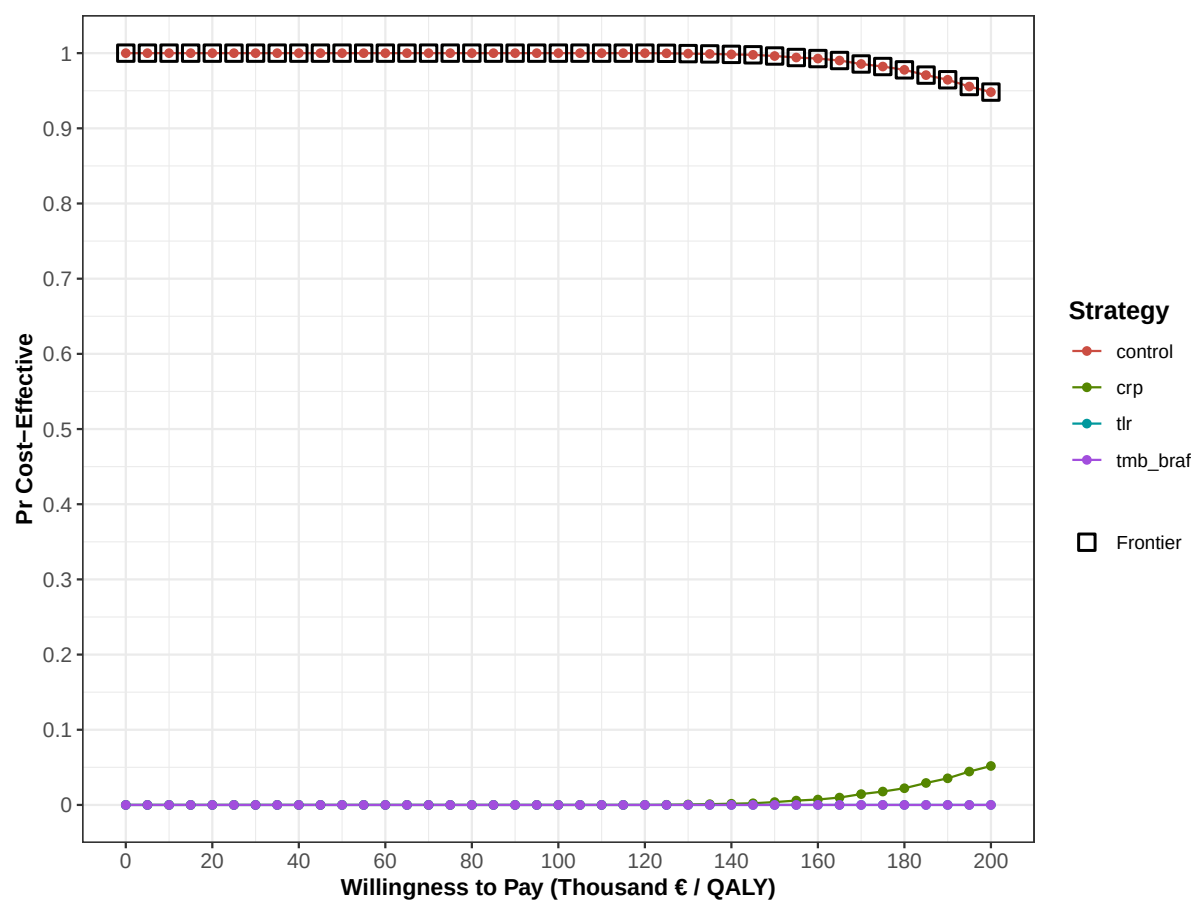


Figure 13: Cost-effectiveness acceptability curves (Scenario 2)

Total EVPI: €0

5 Scenario 3: Higher WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €1e+05

Nivolumab Cost: €4,641 (80% discount)

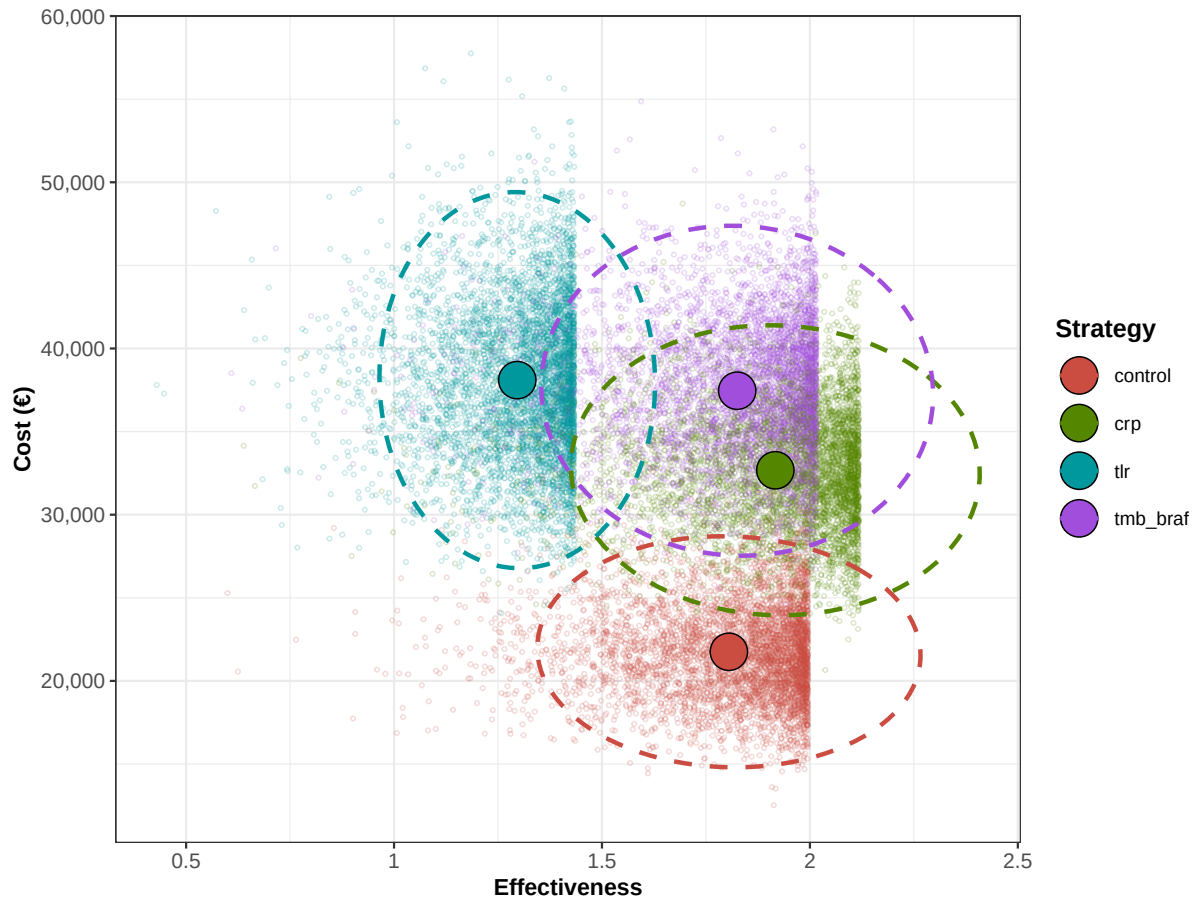


Figure 14: Cost-effectiveness scatter plot (Scenario 3)

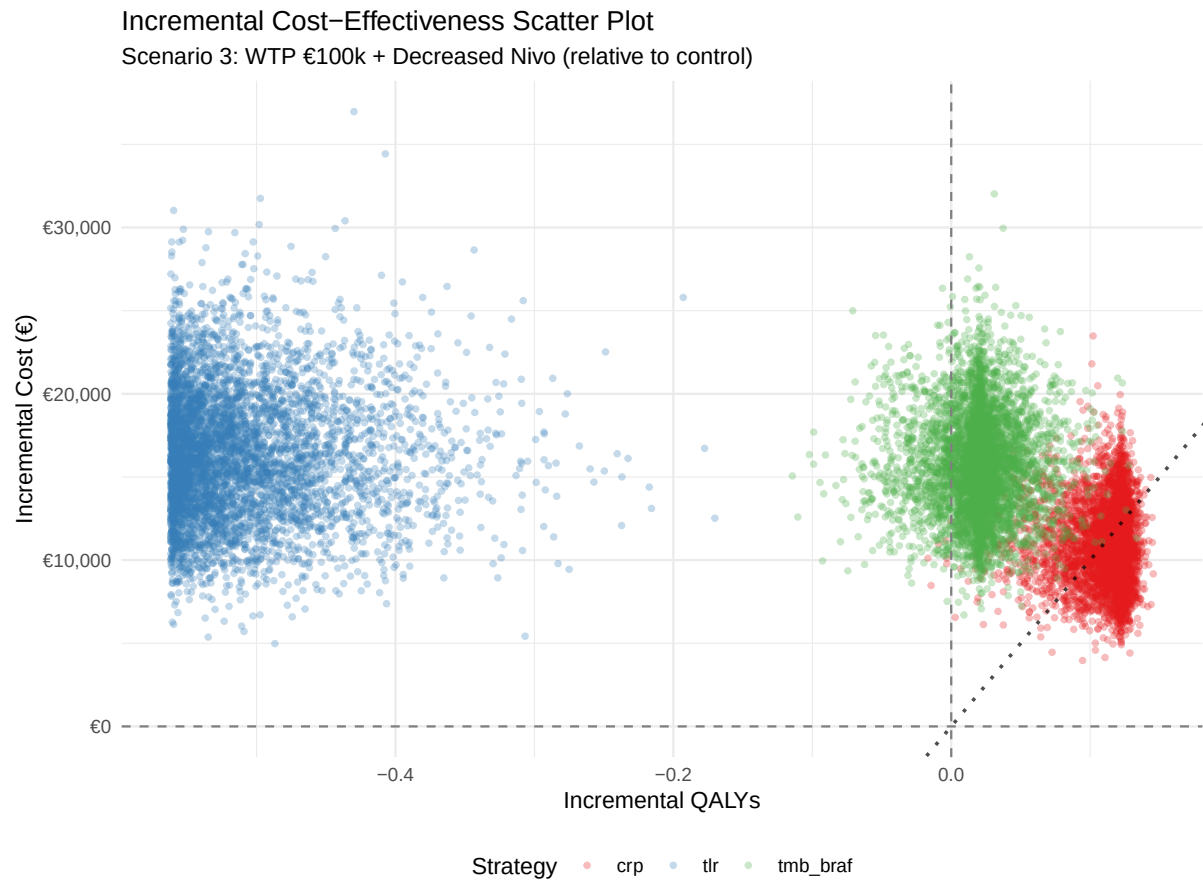


Figure 15: Incremental cost-effectiveness scatter plot relative to control (Scenario 3)

Table 6: Cost-effectiveness analysis summary (Scenario 3)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€21,753.79	€2,839.77	1.805	0.188	€158,768.17	€19,132.96
crp	€32,673.82	€3,564.38	1.917	0.201	€159,011.89	€20,490.60
tlr	€38,096.58	€4,622.78	1.296	0.135	€ 91,534.39	€14,379.14
tmb_braf	€37,458.96	€4,055.46	1.825	0.192	€145,060.77	€19,771.39

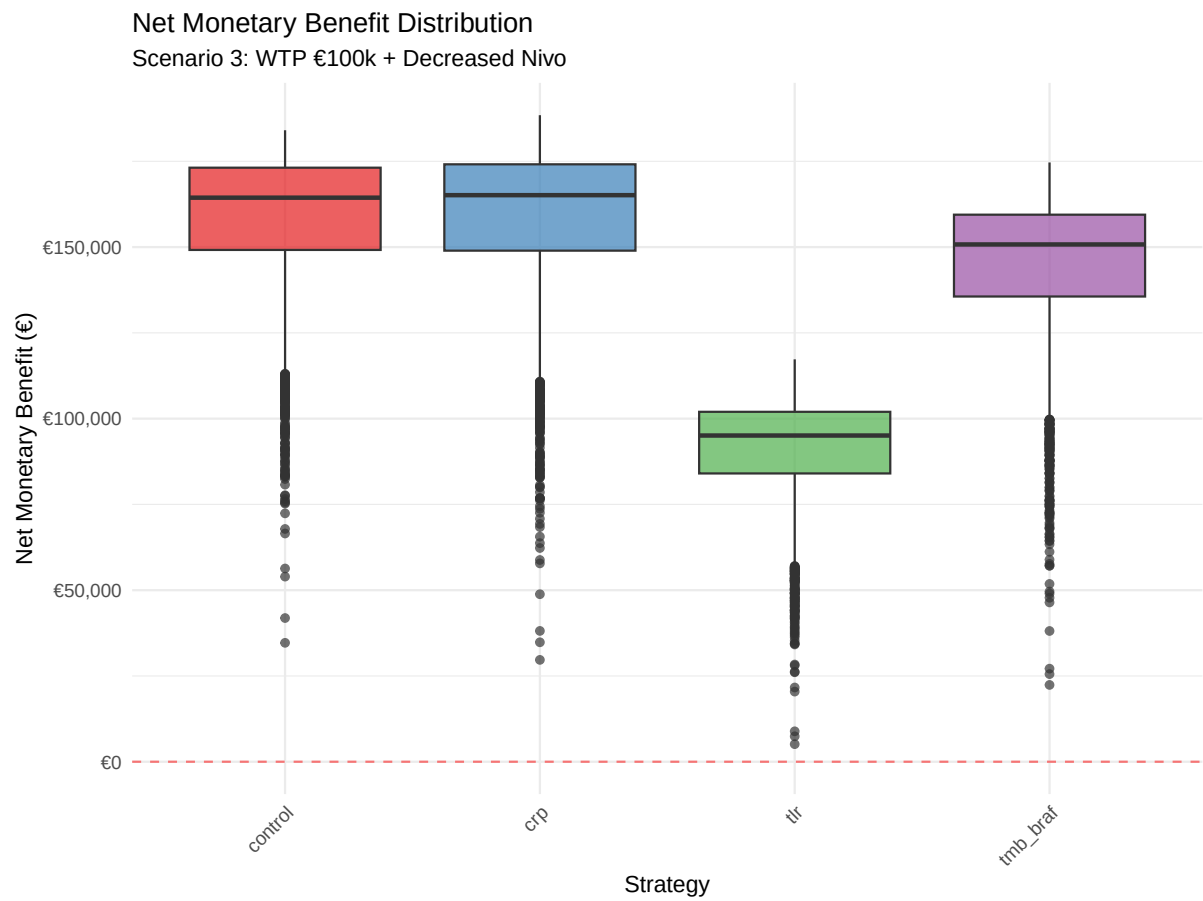


Figure 16: Net Monetary Benefit distribution (Scenario 3)

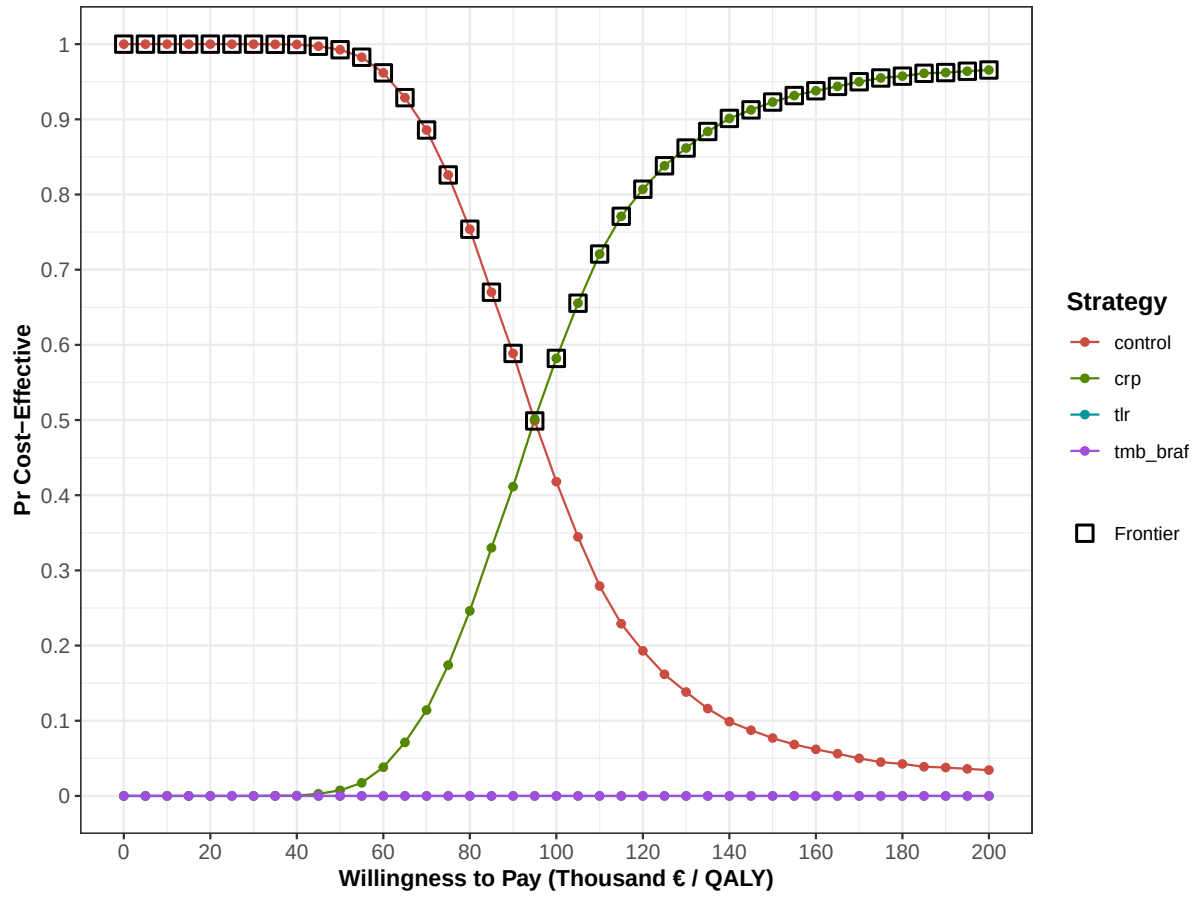


Figure 17: Cost-effectiveness acceptability curves (Scenario 3)

Total EVPI: €1081.33

Table 7: EVPPI results: WTP €100,000 + decreased nivolumab

Parameter	EVPPI	% of EVPI
c_test_CT	€1081.33	100%
u_np	€1081.33	100%
c_other_baseline	€1081.33	100%
c_other_visit	€942.17	87.1%
u_p	€736.93	68.1%
c_drug_nivo	€521.3	48.2%
c_other_follow	€16.59	1.5%
c_drug_FLOX	€0	0%
c_test_NGS	€0	0%
c_other_last	€0	0%

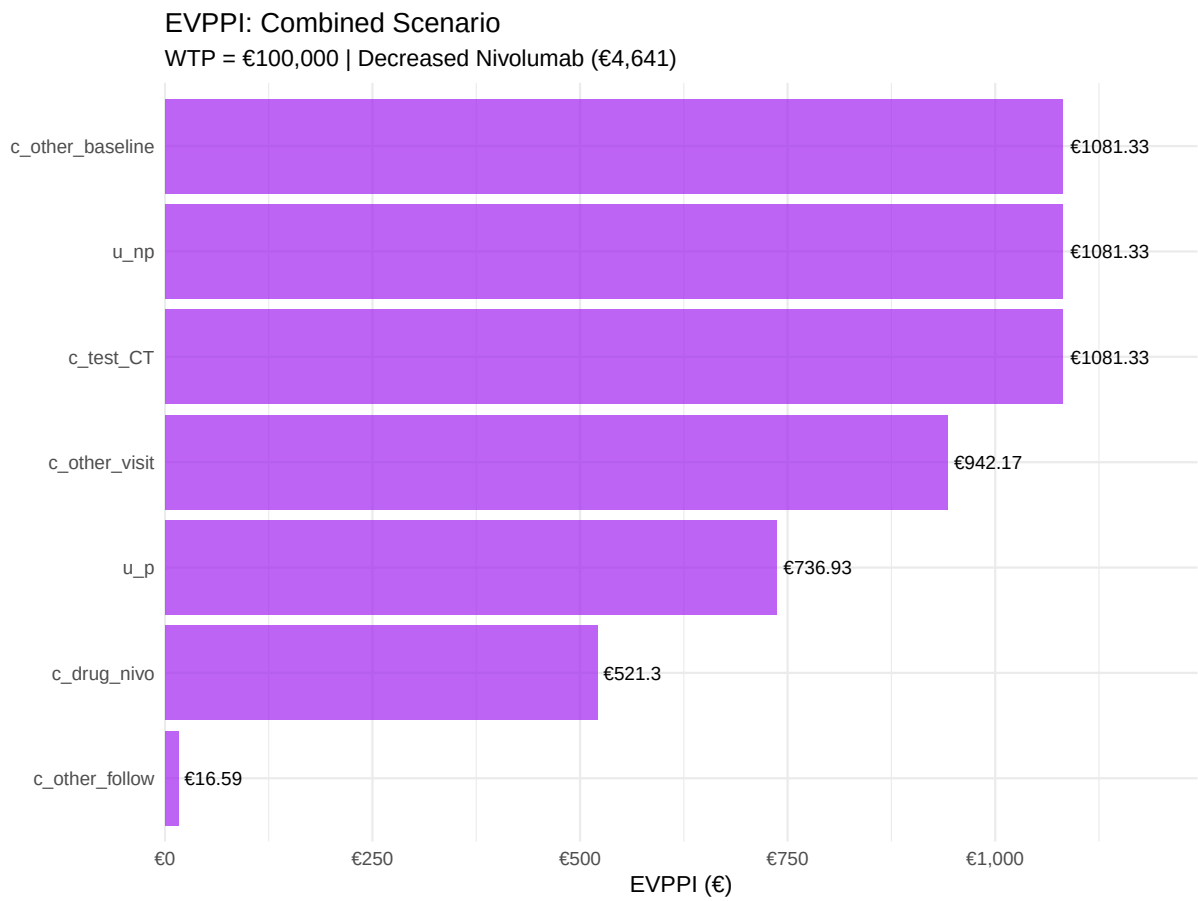


Figure 18: EVPPi: WTP €100,000 with decreased nivolumab

6 Scenario 4: Highest WTP Threshold (€150,000)

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: €13,923 (standard)

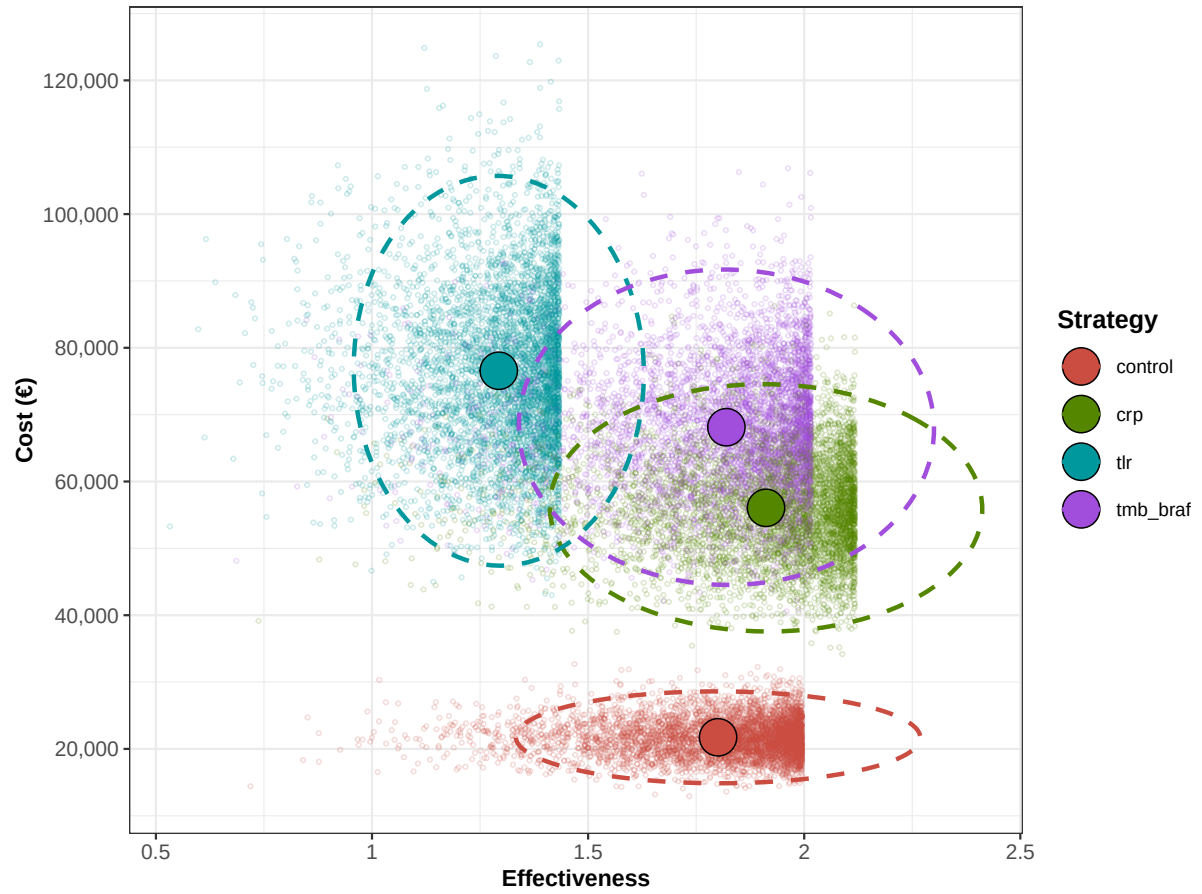


Figure 19: Cost-effectiveness scatter plot (Scenario 4)

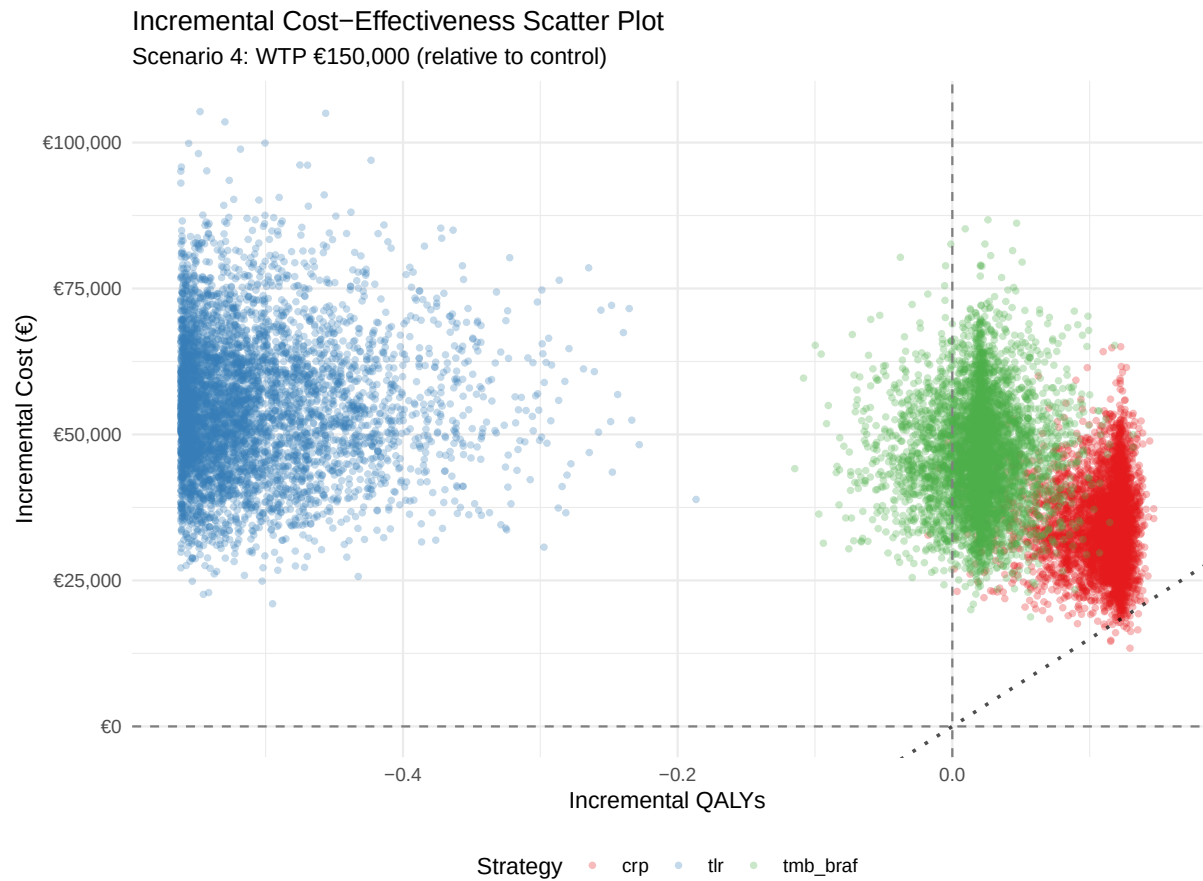


Figure 20: Incremental cost-effectiveness scatter plot relative to control (Scenario 4)

Table 8: Cost-effectiveness analysis summary (Scenario 4)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€21,734.74	€ 2,807.55	1.801	0.191	€248,369.1	€28,829.65
crp	€56,060.40	€ 7,557.54	1.912	0.204	€230,682.1	€31,670.37
tlr	€76,564.22	€11,912.95	1.293	0.137	€117,435.1	€23,914.38
tmb_braf	€68,120.65	€ 9,638.89	1.820	0.196	€204,869.8	€31,075.49

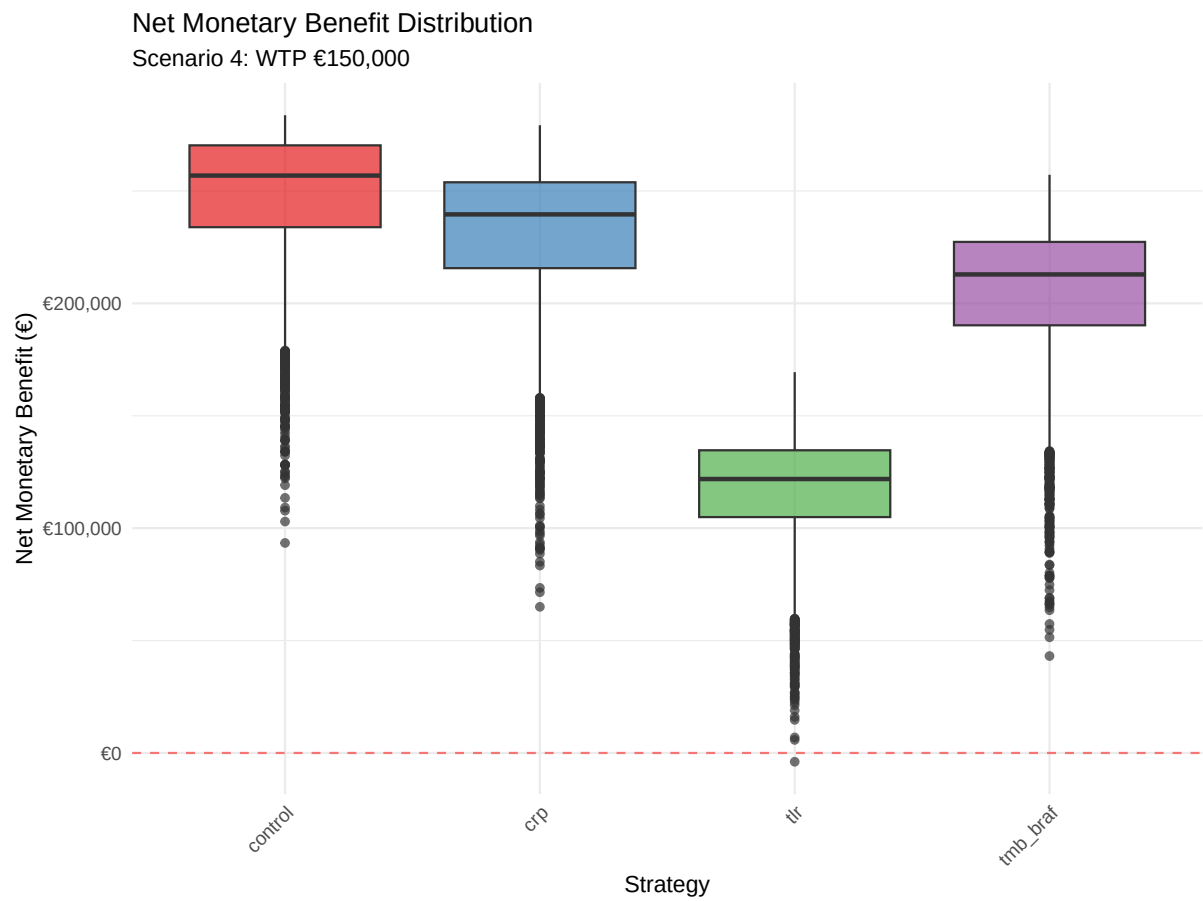


Figure 21: Net Monetary Benefit distribution (Scenario 4)

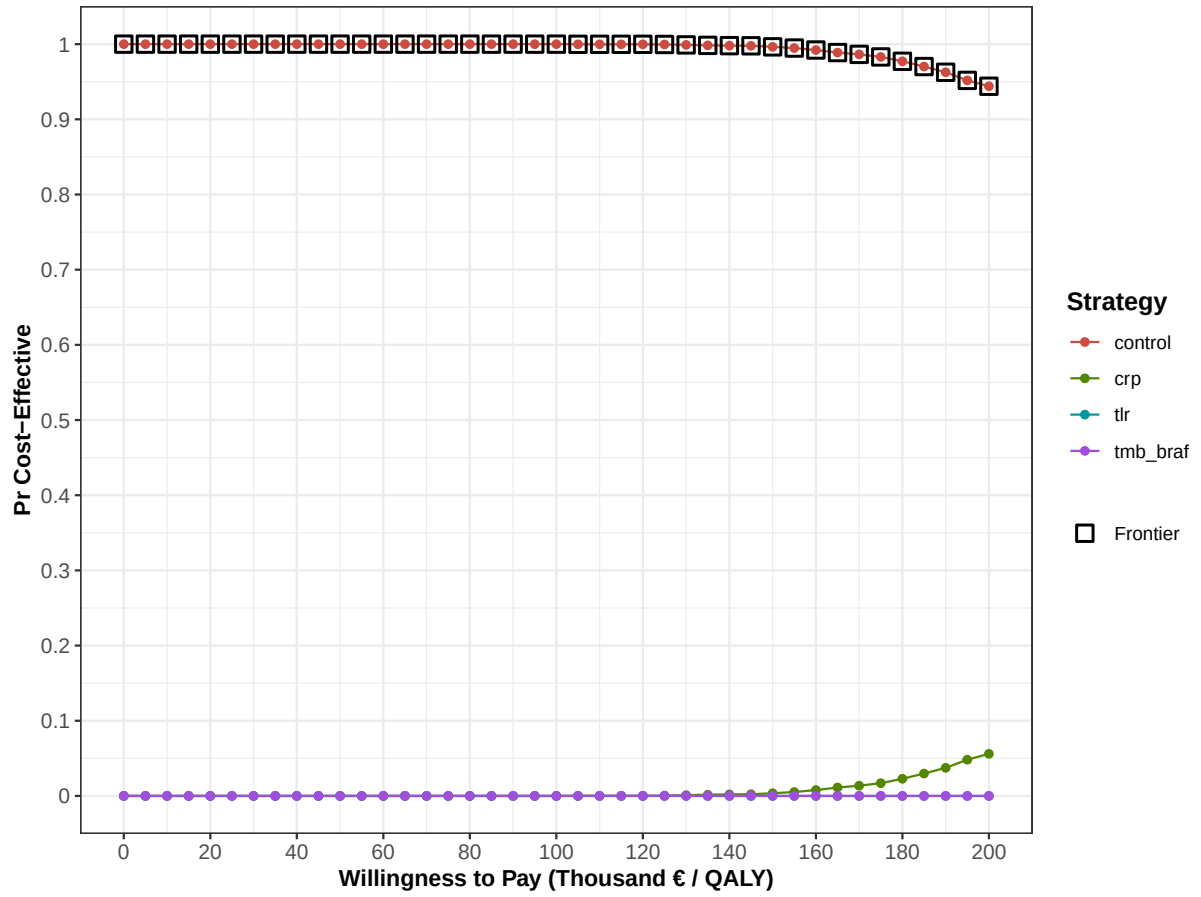


Figure 22: Cost-effectiveness acceptability curves (Scenario 4)

Total EVPI: €6.02

Table 9: EVPPI results: WTP €150,000

Parameter	EVPPI	% of EVPI
c_drug_nivo	€6.02	100%
c_test_NGS	€6.02	100%
c_test_CT	€6.02	100%
u_np	€6.02	100%
u_p	€6.02	100%
c_other_last	€6.02	100%
c_drug_FLOX	€0	0%
c_other_visit	€0	0%
c_other_baseline	€0	0%
c_other_follow	€0	0%

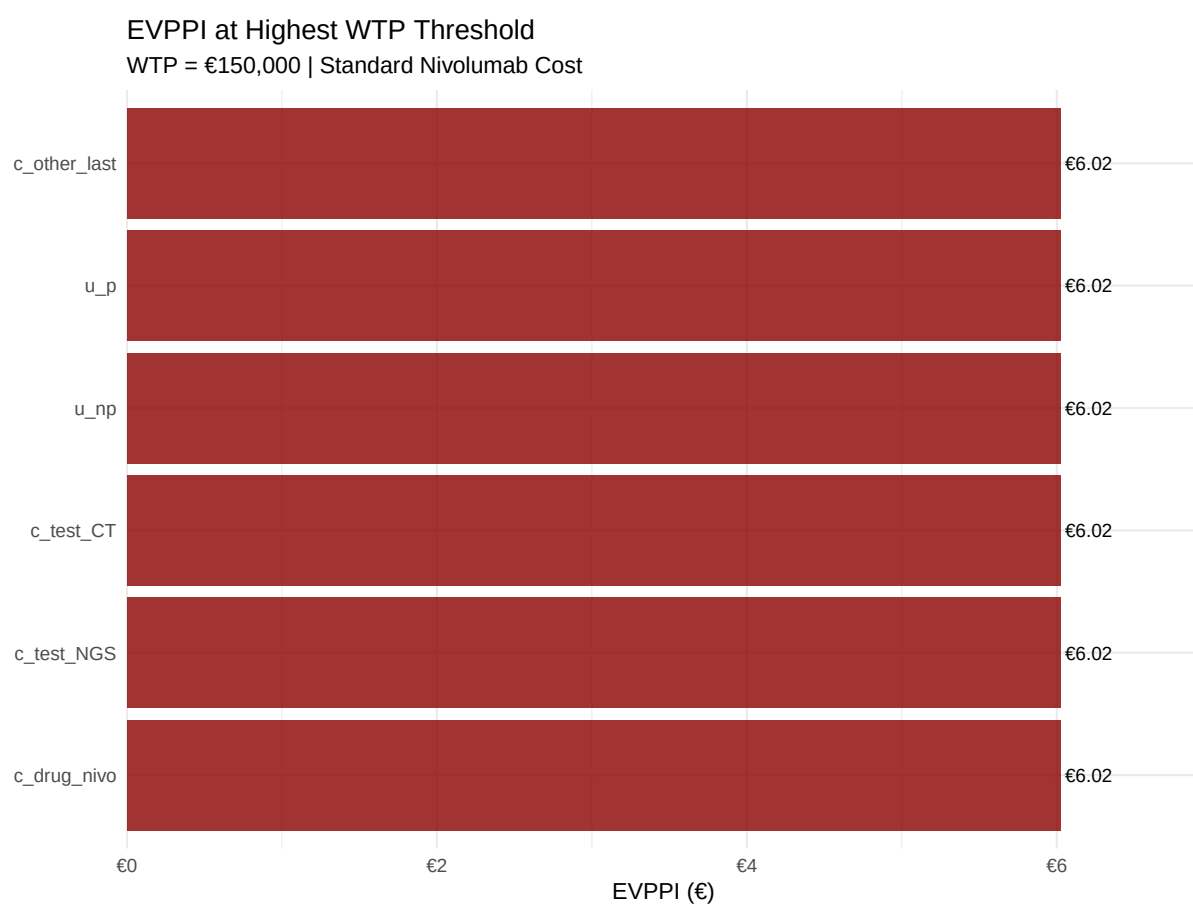


Figure 23: EVPPPI at WTP €150,000

7 Scenario 5: Highest WTP with Decreased Nivolumab Cost

Willingness-to-Pay Threshold: €150,000

Nivolumab Cost: €4,641 (80% discount)

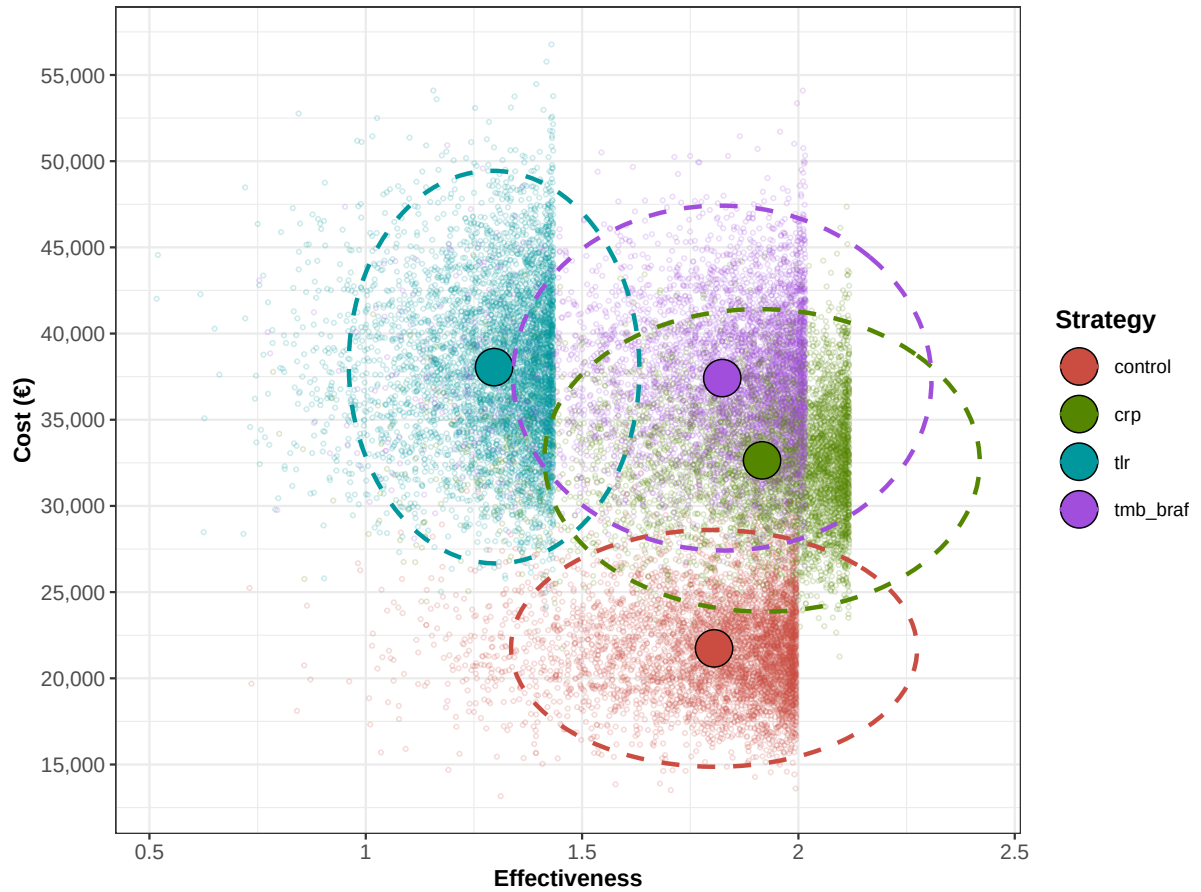


Figure 24: Cost-effectiveness scatter plot (Scenario 5)

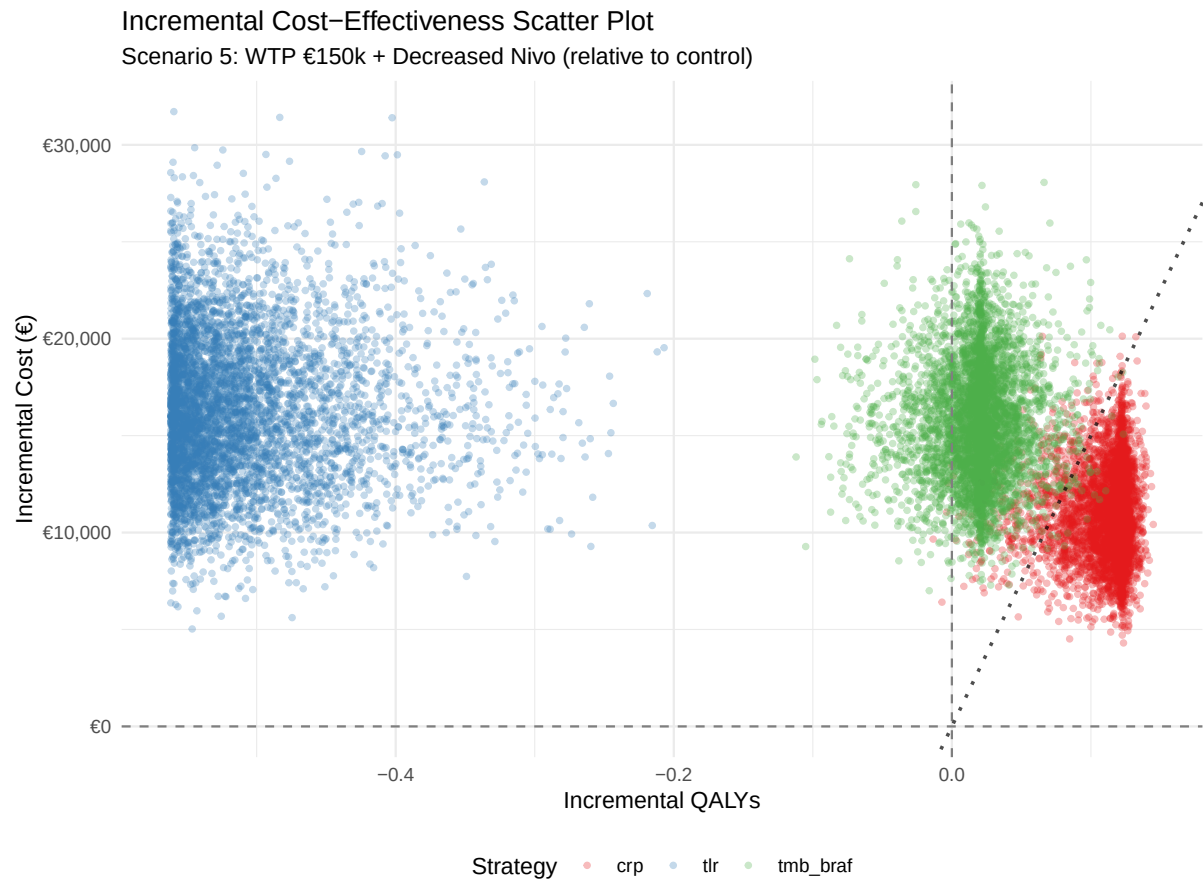


Figure 25: Incremental cost-effectiveness scatter plot relative to control (Scenario 5)

Table 10: Cost-effectiveness analysis summary (Scenario 5)

Strategy	Mean Cost (€)	SD Cost	Mean Effect (QALYs)	SD Effect	Mean NMB (€)	SD NMB
control	€21,736.64	€2,806.95	1.805	0.192	€249,019.8	€28,918.32
crp	€32,643.39	€3,585.45	1.916	0.206	€254,756.9	€31,084.54
tlr	€38,052.81	€4,653.81	1.297	0.137	€156,437.1	€21,166.44
tmb_braf	€37,416.03	€4,088.11	1.824	0.198	€236,178.8	€29,967.19

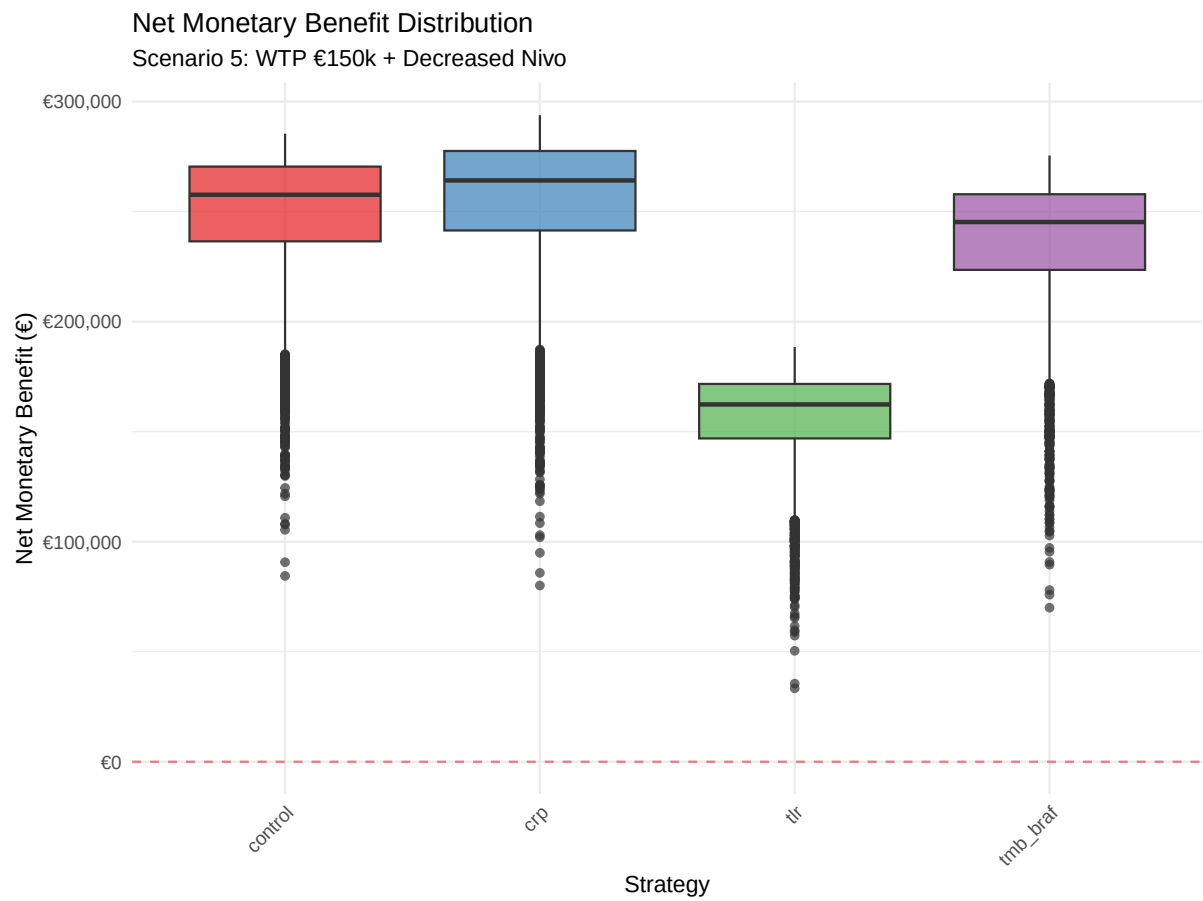


Figure 26: Net Monetary Benefit distribution (Scenario 5)

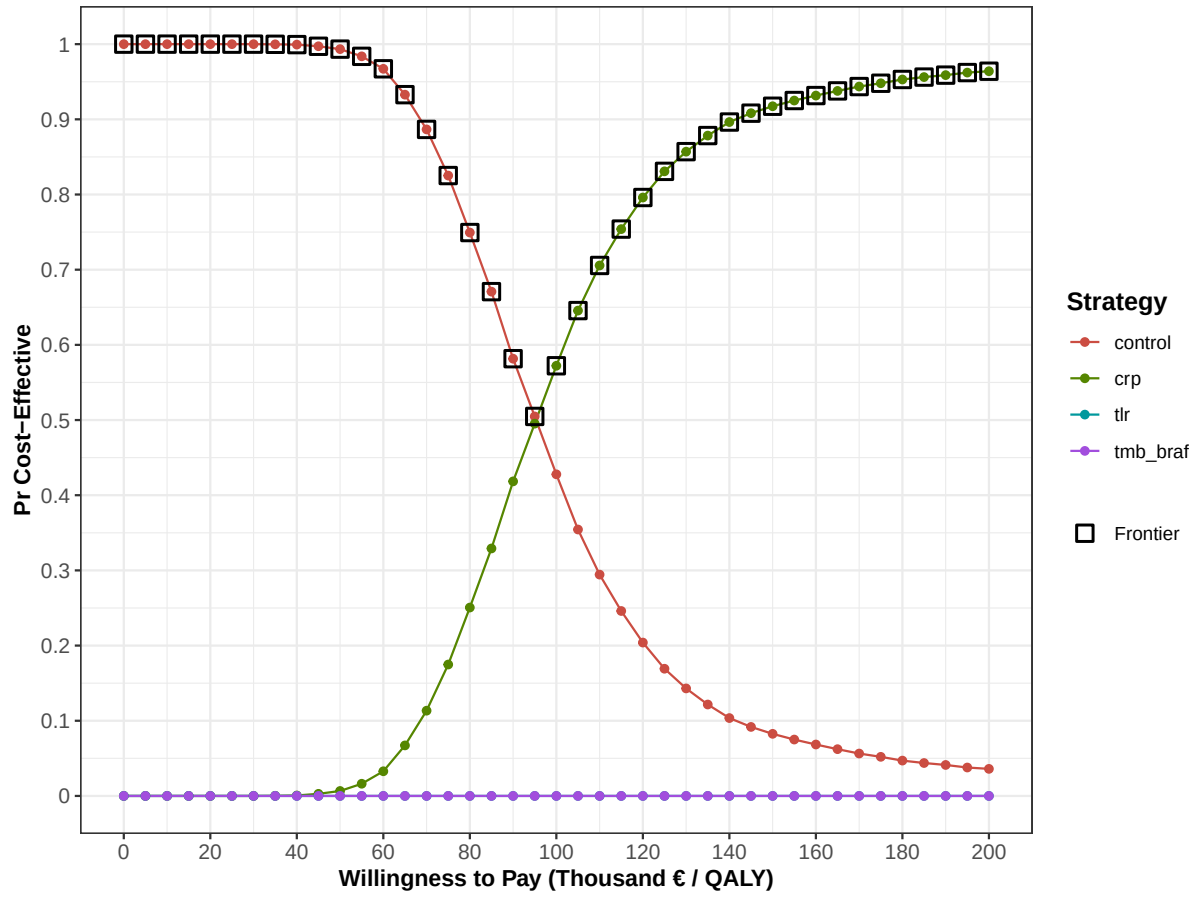


Figure 27: Cost-effectiveness acceptability curves (Scenario 5)

Total EVPI: €273.11

Table 11: EVPPI results: WTP €150,000 + decreased nivolumab

Parameter	EVPI	% of EVPI
c_drug_nivo	€273.11	100%
c_drug_FLOX	€273.11	100%
c_test_NGS	€273.11	100%
u_np	€273.11	100%
u_p	€273.11	100%
c_other_last	€273.11	100%
c_other_baseline	€273.11	100%
c_test_CT	€0	0%
c_other_visit	€0	0%
c_other_follow	€0	0%

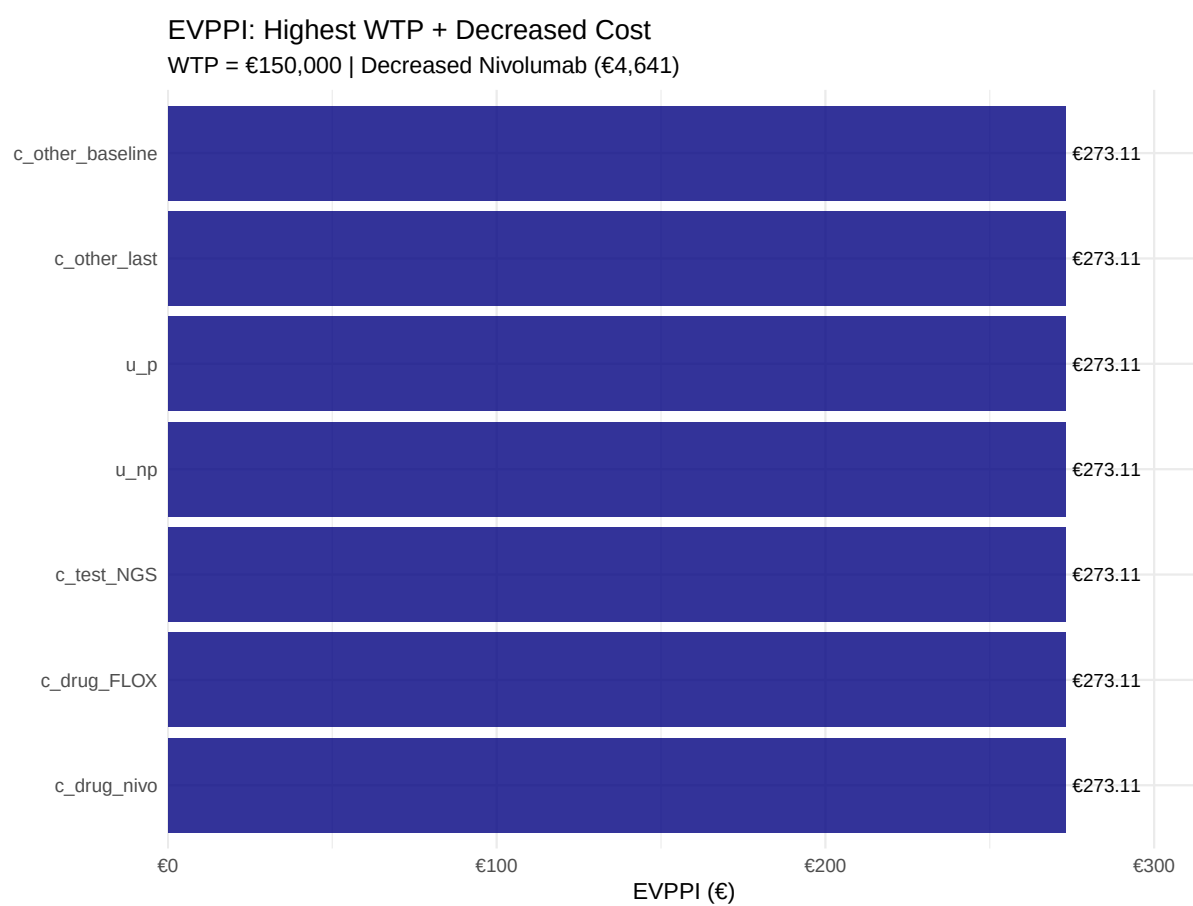


Figure 28: EVPPPI: Highest WTP with decreased nivolumab cost

8 Comparative Analysis Across Scenarios

8.1 EVPI



Figure 29: Expected Value of Perfect Information across scenarios

8.2 EVPPI

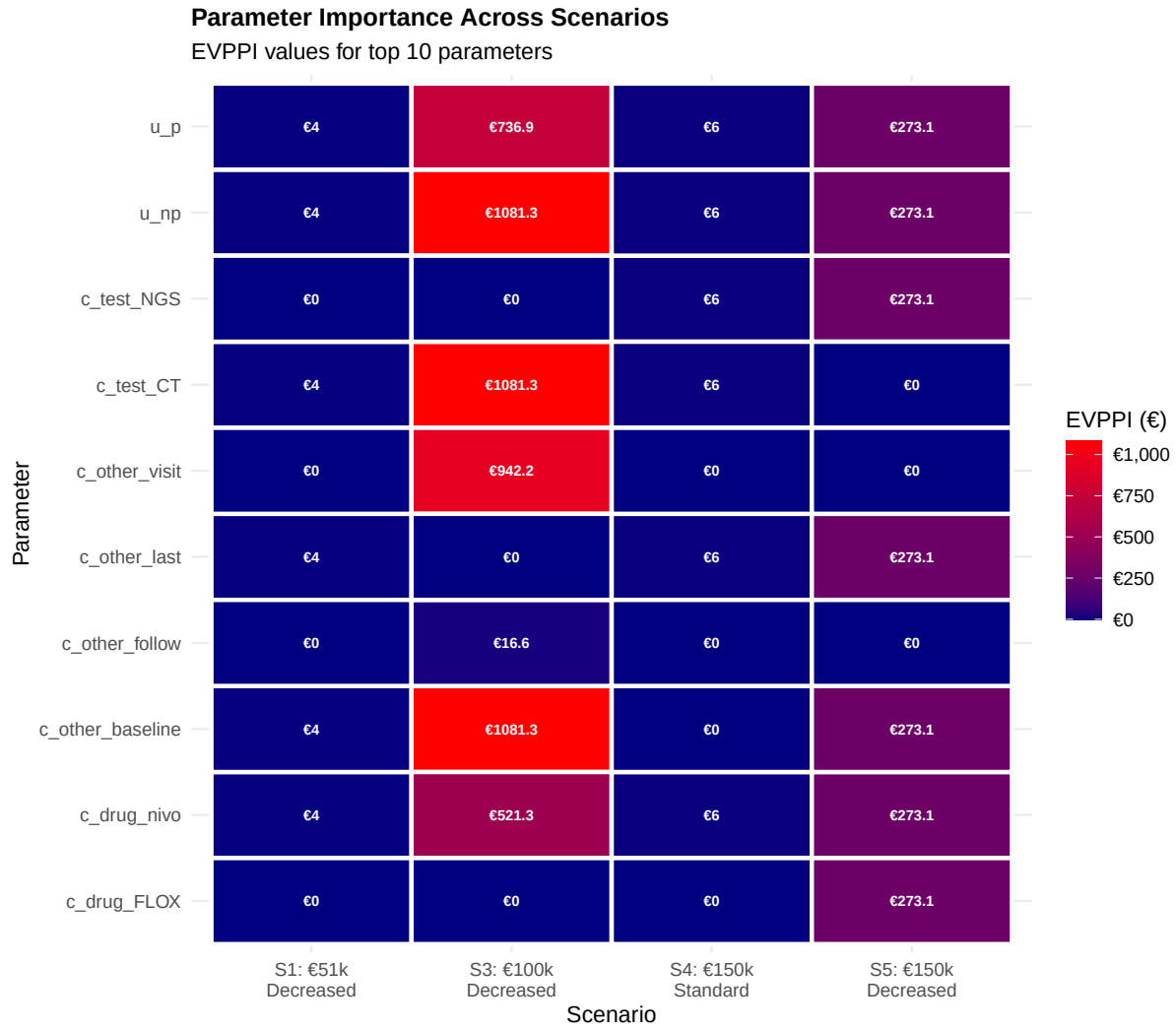


Figure 30: Parameter importance across scenarios (top 10 parameters)

8.3 Summary

Table 12: Summary of EVPI and top EVPPI by scenario

Scenario	WTP	EVPI	Top Parameter	Max EVPPI	% of EVPI
Decreased Nivolumab Cost	€ 51,000	€4.01	c_drug_nivo	€4.01	100
WTP €100,000 + Decreased Nivolumab	€100,000	€1081.33	c_test_CT	€1081.33	100
WTP €150,000	€150,000	€6.02	c_drug_nivo	€6.02	100
WTP €150,000 + Decreased Nivolumab	€150,000	€273.11	c_drug_nivo	€273.11	100

Scenario with Greatest Decision Uncertainty: WTP €100,000 + Decreased Nivolumab (EVPI = €1081.33)

Most Important Parameter Across All Scenarios: u_np (Total EVPPI = €1364.48)

Impact of WTP Threshold: Higher WTP thresholds generally increase decision uncertainty (EVPI), creating greater value for additional research to reduce parameter uncertainty.

Impact of Nivolumab Cost: Decreased nivolumab costs shift the cost-effectiveness landscape and may change which parameters drive decision uncertainty.